UGANDA CHRISTIAN UNIVERSITY
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ELE1204: Digital Electronics and Logic Design

Lecture 05 (BSCS)

Flip-Flops: SR, JK, D, T

Ms. Olivia Nakayima
(MSc, BSc. CCNA, JNCIA, JNCIS)
Department of Computing & Technology
Faculty of Engineering, Design & Technology

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Lecture Objectives and Learning outcomes

The Objectives of this lecture are :

- To understand the sequential memory elements
- To analyse them and their applications

By the end of this lecture, students should be able to:

- Define sequential memory elements
- Fully explain the SR, JK, D, T Flip Flops

NOTE:: SMART = **S**pecific, **M**easurable, **A**chievable, **R**elevant, **T**imebound

Lecture Overview

- Introduction to Flip-Flops
- Types of Flip-Flops
- SR (Set-Reset) Flip-Flop
- **JK Flip-Flop** (Modified SR Flip-Flop)
- D (Data or Delay) , Toggle Flip-Flops

Introduction to Flip-Flops

- Digital systems can operate either asynchronously or synchronously. In asynchronous systems, the outputs of logic circuits can change state any time one or more of the inputs change.

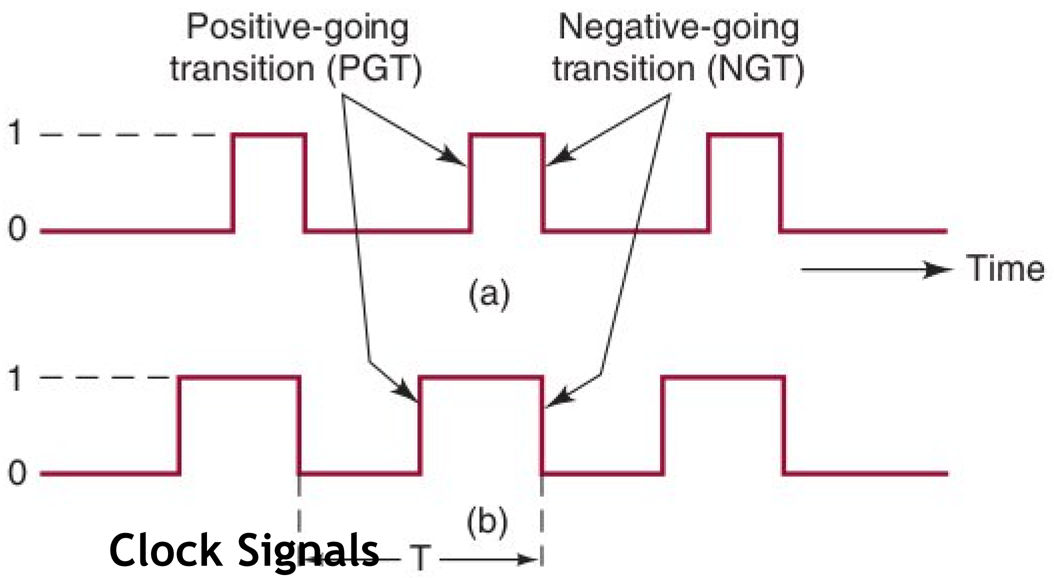
- An asynchronous system is generally more difficult to design and troubleshoot than a synchronous system.
- In synchronous systems, the exact times at which any output can change states are determined by a signal commonly called the **clock.**
- **System outputs can change state only when the clock makes a transition.**
- When the clock changes from a 0 to a 1, this is called the **positive-going transition (PgT);** when the clock goes from 1 to 0, this is the **negative-going transition (NgT).**

The synchronizing action of the clock signals is accomplished through the use of

Introduction to Flip-Flops

- Flip-flops are **sequential logic circuits** used for **data storage, synchronization, and state transitions** in digital systems.

- Unlike combinational circuits, flip-flops have **memory**, meaning their output depends on BOTH **current inputs and past states**.
- https://www.youtube.com/watch?v=LTtuYeSm



Clock Signals

Introduction to Flip-Flops

- A clock cycle is measured from one PGT to the next PGT or from one NGT to the next NGT.
- The time it takes to complete one cycle (seconds/cycle) is called

  the period (T).

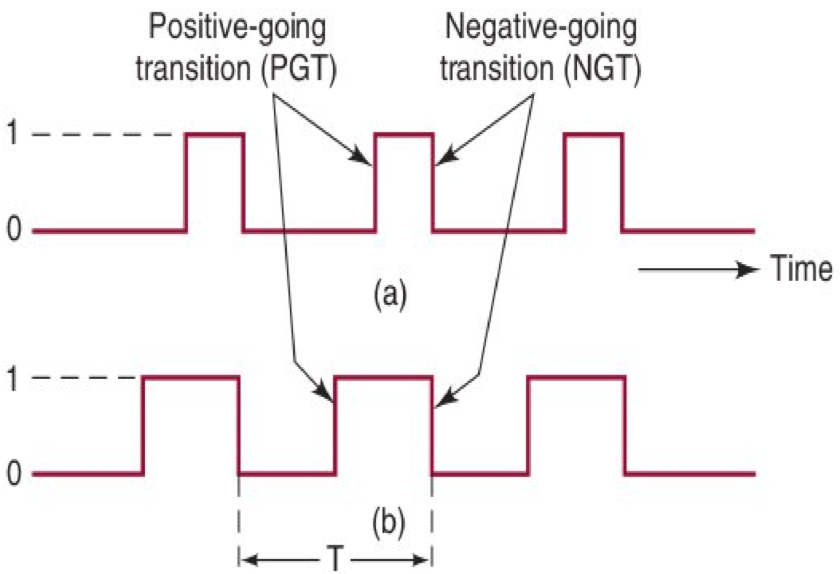
- The speed of a digital system is normally referred to by the number of cl

  that happen in 1 s (cycles/second),

  which is known as the frequency (f) of the clock.

- The standard unit for frequency is hertz. One

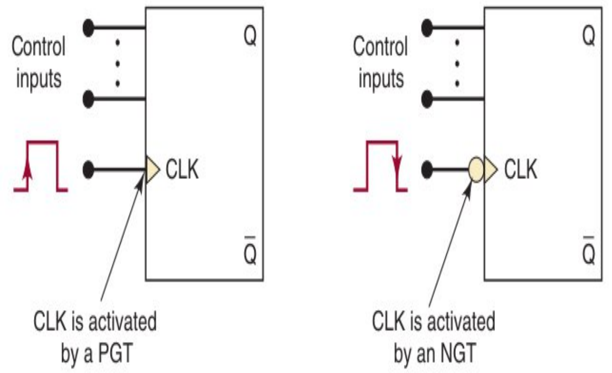
  hertz 11 Hz2 = 1 cycle/second.



Introduction to Flip-Flops

Clocked Flip-Flops

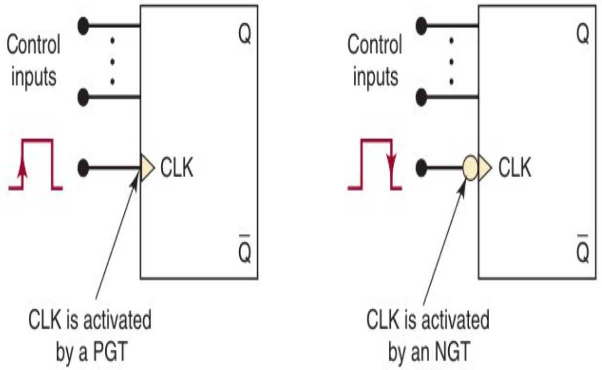
- Clocked FFs have a clock input that is typically labeled CLK, CK, or CP (clock pulse).
- The CLK input is **edge-triggered**, which means that it is activated by a signal transition; this is indicated by the presence of a small triangle on the **CLK** input.
- Clocked FFs also have **one or more control inputs** that can have various names, depending on their operation the control inputs get the FF outputs ready to change, while the active transition at the CLK input actually triggers the change.



Introduction to Flip-Flops

Clocked Flip-Flops

- The control inputs control the **WHAT** (i.e., what state the output will go to); the **CLK** input determines the **WHEN.**
- The control inputs will have no effect on **Q** until the active clock transition occurs. In other words, their effect is synchronized with the signal applied to CLK. For this reason they are called **synchronous control inputs**.



Introduction to Flip-Flops

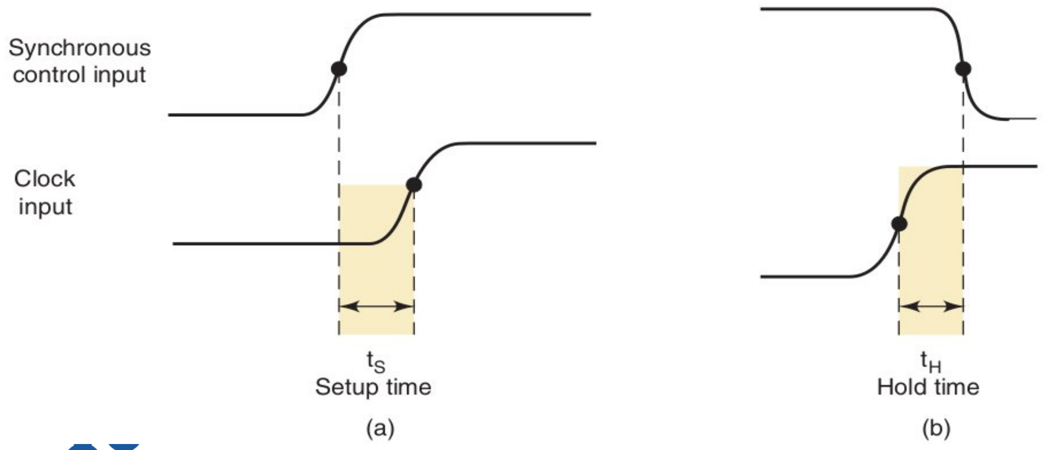
• Set up and Hold Times

Setup time is the **minimum amount of time** before the **clock signal changes** (rising or falling edge) that the **input signal (data)** must be stable (not changing) so that the flip-flop or latch can correctly store the data.

Hold time is the **minimum amount of time** that the **input signal (data)** must remain stable **after** the clock edge to ensure the flip-flop correctly stores the data.

Flip-flop will perform in an unacceptable way before settling to either HIGH or LOW.

Short-term abnormal voltages present on the output that are referred to as **metastable states.**



Aspect	Setup Time	Hold Time
When it happens?	**Before** the clock edge	**After** the clock edge
Purpose	Ensures data is stable **before clock triggers**	Ensures data remains stable **after clock triggers**
What happens if violated?	Flip-flop may store **wrong or unstable data**	Flip-flop may enter **metastable state**

Types of Flip-Flops

SR (Set-Reset) Flip-Flop

- Stores a **single bit** of data using **Set (S)** and **Reset (R)** inputs.
  - **https://www.youtube.com/watch?v=HZg7fNu-I24**

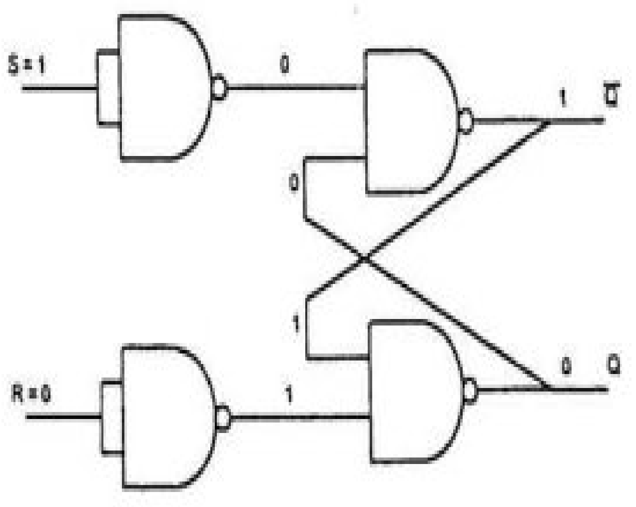
Truth table

S	R	Q(Next state)	Q’ Complement
0	0	No change	No change
0	1	0	1
1	0	1	0
1	1	Invalid	Invalid

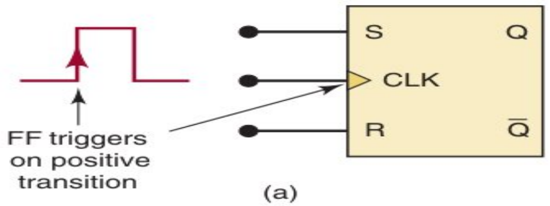
S = 1, R = 0 → Sets the output to **1**.
S = 0, R = 1 → Resets the output to **0**.
S = 0, R = 0 → Holds the previous state.
S = 1, R = 1 → **Invalid condition**, causes unpredictable behavior.

Types of Flip-Flops

SR (Set-Reset) Flip-Flop



SR FLIP-FLOP

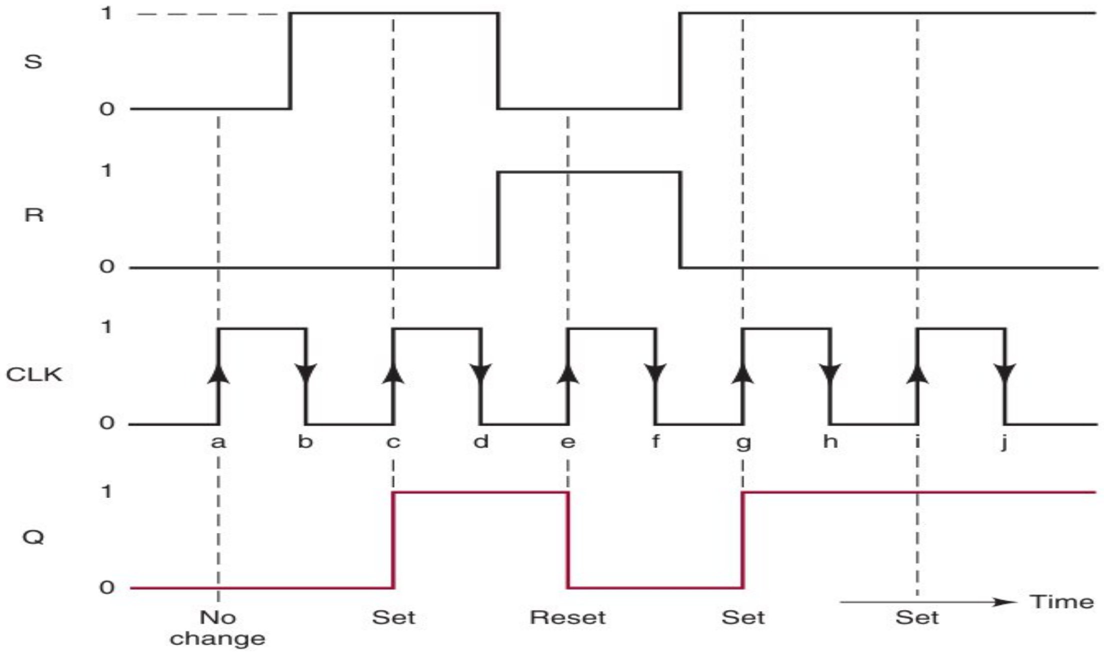


(a)

Inputs			Output
S	R	CLK	Q
0	0	↑	Q_0 (no change)
1	0	↑	1
0	1	↑	0
1	1	↑	Ambiguous

Q_0 is output level prior to ↑ of CLK.
↓ of CLK produces no change in Q.

(b)



Types of Flip Flops

JK Flip-Flop (Modified SR Flip-Flop)

- Eliminates the **invalid state** of the SR flip-flop by toggling when **J = K = 1**.
  - **https://www.youtube.com/watch?v=j6krFp511HA**

Truth table

J	K	Q(Next state)	Q’ Complement
0	0	No change	No change
0	1	0	1
1	0	1	0
1	1	Toggle (Q-Q’)	Toggle

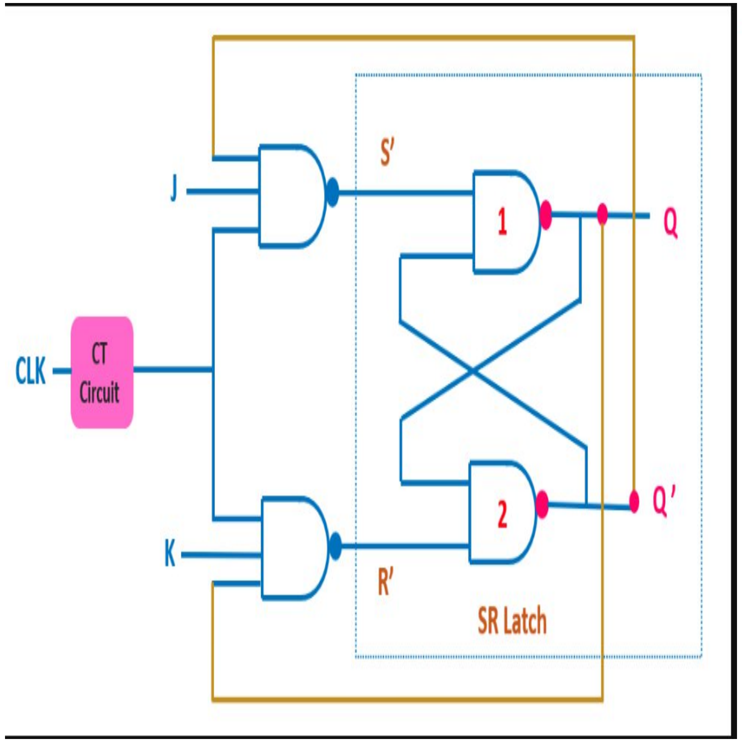
J = 1, K = 0 → Sets the flip-flop (Q = 1).
J = 0, K = 1 → Resets the flip-flop (Q = 0).
J = 0, K = 0 → Holds the previous state.
J = 1, K = 1 → Toggles the state (Q → Q').

Usage: Counters, shift registers, frequency dividers.

Types of Flip Flops

JK Flip-Flop (Modified SR Flip-Flop)

- The J and K inputs control the state of the FF in the same ways as the S and R inputs do for the clocked S-R flip-flop except for one major difference: the J = K = 1 condition does not result in an ambiguous output.
- When J and K are both 1, the FF will always go to its opposite state upon the positive transition of the clock signal. This is called the **toggle mode** of operation. In this mode, if both J and K are left HIGH, the FF will change states (toggle) for each PGT of the clock.



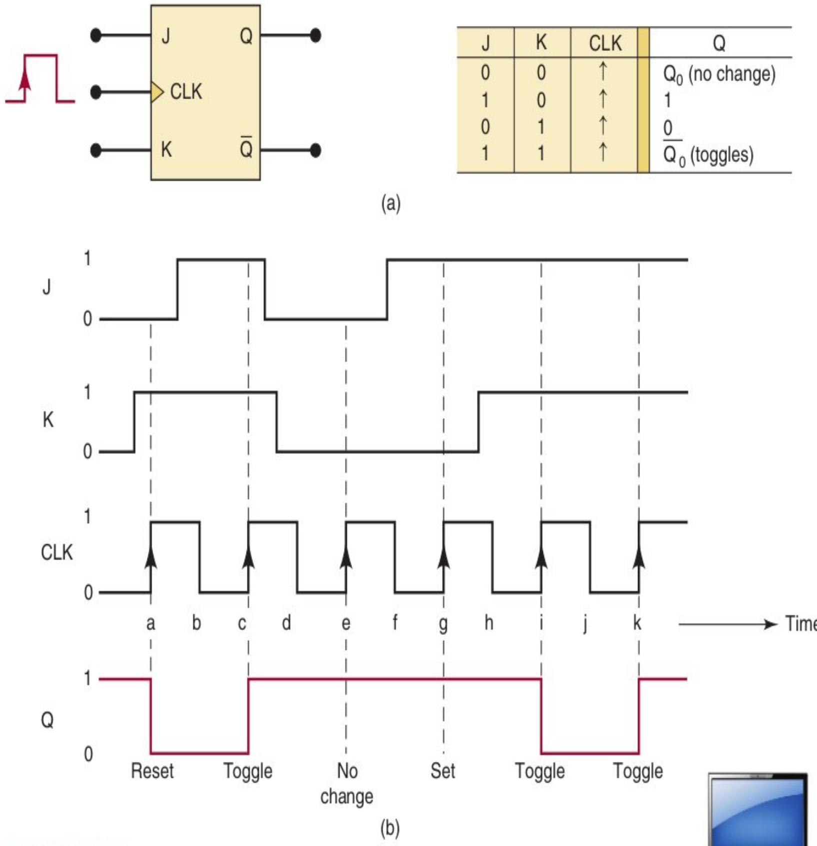


FIGURE 5-24 (a) Clocked J-K flip-flop that responds only to the positive edge of the clock; (b) waveforms.

Types of Flip Flops

D (Data or Delay) Flip-Flop

- Unlike the S-R and J-K flip-flops, this flip-flop has only one synchronous control input, D, which stands for data: **https://www.youtube.com/watch?v=dnfXXpW7tlw**
- Stores **one bit of data**; output follows the **D input** at the clock edge.

D	CLK	Q(Next state)	Q’ Complement
0	↑	0	1
1	↑	1	0

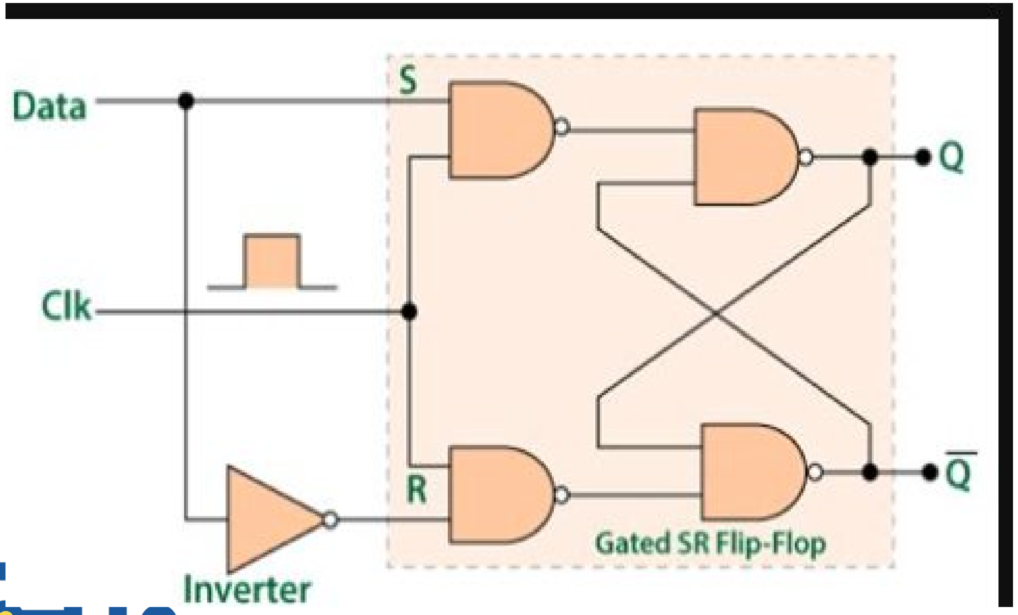
Usage: Used in registers, memory, and synchronization circuits.

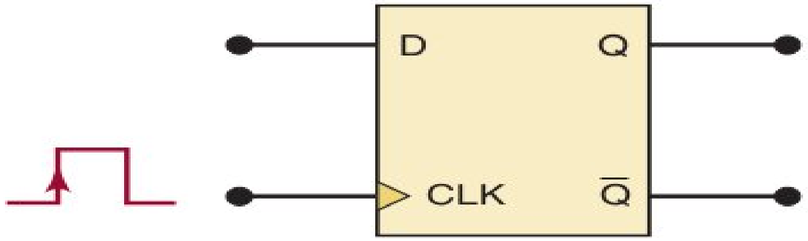
- Acts like a **latch**, capturing input **D** only when the **clock (CLK) is HIGH**.
- Prevents **invalid states** like in **SR Flip-Flop**.
- The level present at D will be stored in the flip-flop at the instant the PGT occurs.

Types of Flip Flops

D (Data or Delay) Flip-Flop

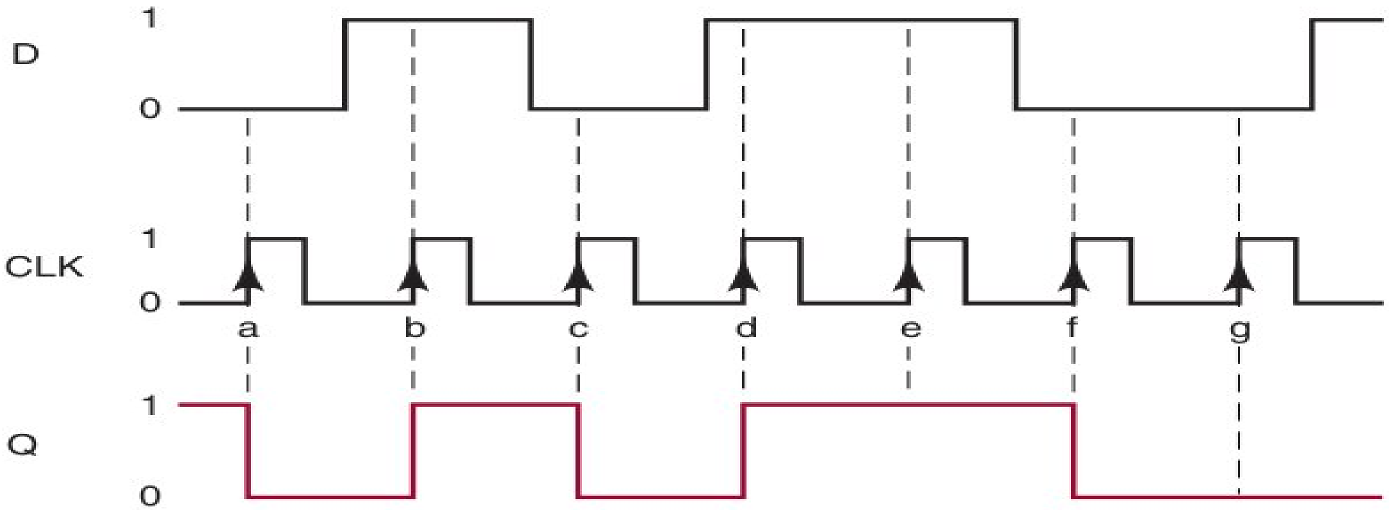
- **Q** will go to the same state that is present on the D input when a PGT occurs at CLK





D	CLK	Q
0	↑	0
1	↑	1

(a)



(b)

Types of Flip Flops

T (Toggle) Flip-Flop

- **Toggles (flips) the output state** whenever **T = 1 and the clock signal changes**.

Truth table

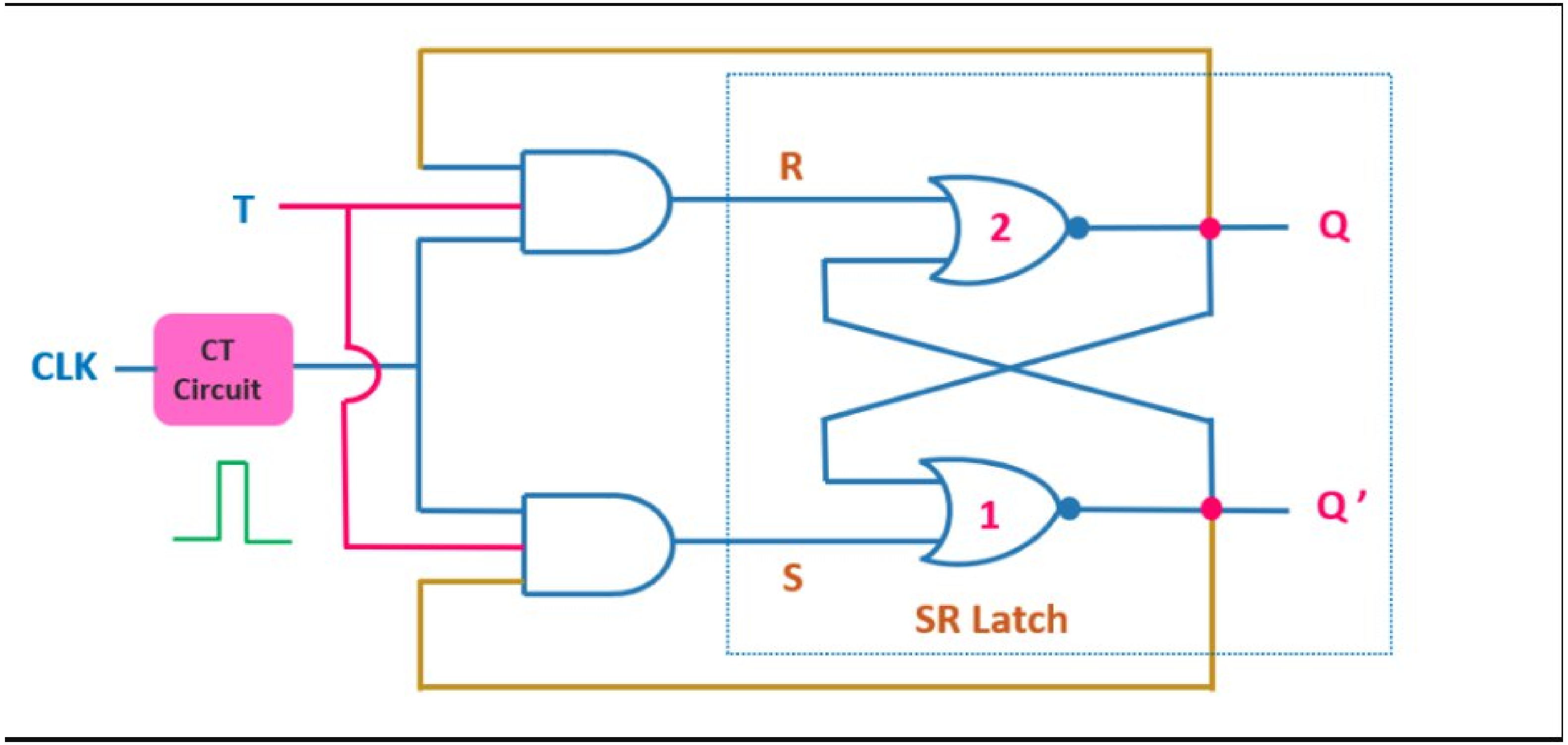
T	CLK	Q(Next state)	Q' Complement
0	↑	No change	No change
1	↑	Toggle	Toggle

- If **T = 0**, it **maintains the previous state**.
- If **T = 1**, it **toggles the output** at every clock pulse.

Usage: Counters, frequency dividers, toggling circuits.

Types of Flip Flops

T (Toggle) Flip-Flop



Types of Flip Flops

Applications of Flip-Flops

- **Memory Storage:** Flip-flops store **1-bit data** in registers, caches, and RAM.
- **Frequency Dividers: T Flip-Flops** divide clock signals in digital clocks and microcontrollers.
- **State Machines: JK & D Flip-Flops** are used in FSM (Finite State Machines) for **control systems**.
- **Shift Registers: D & JK Flip-Flops** are used for **serial-to-parallel conversion** in communication systems.

Types of Flip Flops

Comparison of Flip-Flops

Feature	SR	JK	D	T
Basic Function	Set - Reset	Enhanced SR	Data Storage	Toggle
Invalid State	Yes (S=1, R=1)	No	No	No
Toggle capability	No	Yes	No	Yes
Best Use case	Basic Storage	Counter	Registers	Frequency Division

Thank You

Uganda Christian University
P.O. Box 4 Mukono, Uganda
Tel: 256-312-350800
https://ucu.ac.ug/ Email: info@ucu.ac.ug.
@ugandachristianuniversity @UCUniversity
@UgandaChristianUniversity

Department of Computing & Technology
FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY
Tel: +256 (0) 312 350 863 | WhatsApp: +256 (0) 708 114 300
@ucucomputeng @ucu_ComputEng
https://cse.ucu.ac.ug/ Email: dct-info@ucu.ac.ug